

Server reference data



Labeled examples

Class-balanced
sampling

Hidden tasks

Server-only



Weakest-task utility

Rank penalty

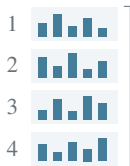
Previous weights



Client weights

cap

Client updates

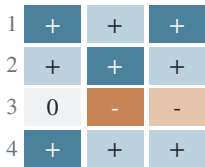


Robust
clipping

Scoring copies

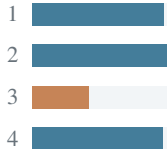


Equal scale



Task-wise loss reduction

■ Improves ■ Harms



Clipped updates

Weighted
aggregation

Updated
model

